more hallucinations with multiple ways to state each fact. In the case of fixed-format birthdays, $|\mathcal{E}_c| = 364$ and notable people whose birthdays are discussed often would have high $\mu(c)$. Notable birthdays like Einstein's appear multiple times, whereas others may only occur once, e.g., in an obituary. Large language models seldom err on frequently referenced facts, e.g., Einstein's birthday or dissertation title.

Our lower-bound for hallucinations is based on the fraction of prompts appearing just once in the training data, ignoring IDKs.

Definition 2 (Singleton rate)**.** *A prompt $c \in \mathcal{C}$ is a* singleton *if it appears exactly once in the N training data $\langle (c^{(i)}, r^{(i)}) \rangle_{i=1}^{N}$ without abstention, i.e., $|\{i : c^{(i)} = c \wedge r^{(i)} \neq \text{IDK}\}| = 1$. Let $\mathcal{S} \subseteq \mathcal{C}$ denote the set of singletons and*

$$\text{sr} = \frac{|\mathcal{S}|}{N}$$

denote the fraction of training singletons.

The singleton rate builds on Alan Turing's elegant "missing-mass" estimator (Good, 1953), which gauges how much probability is still assigned to outcomes that have not yet appeared in a sample from a distribution. Concretely, Turing's estimate of the unseen-event probability is the fraction of samples appearing exactly once. Intuitively, singletons act as a proxy for how many more novel outcomes you might encounter in further sampling, so their empirical share becomes the estimate for the entire "missing" portion of the distribution. We now state our bounds for Arbitrary Facts.

Theorem 2 (Arbitrary Facts)**.** *In the Arbitrary Facts model, any algorithm which takes N training samples and outputs $\hat{p}$ satisfies, with probability $\geq 99\%$ over $\vec{a} = \langle a_c \rangle_{c \in \mathcal{C}}$ and the N training examples:*

$$\text{err} \geq \text{sr} - \frac{2}{\min_c |\mathcal{E}_c|} - \frac{35 + 6 \ln N}{\sqrt{N}} - \delta.$$

Moreover, there is an efficient algorithm outputting calibrated $\hat{p}$ ($\delta = 0$) that with probability $\geq 99\%$,

$$\text{err} \leq \text{sr} - \frac{\text{sr}}{\max_c |\mathcal{E}_c| + 1} + \frac{13}{\sqrt{N}}.$$

An earlier version of this paper presented a related theorem that omitted prompts and abstentions (Kalai and Vempala, 2024). The proof is in Section B. Follow-up work by Miao and Kearns (2025) provides an empirical study of hallucinations, singleton rate, and calibration.

3.3.2 Poor models

Misclassifications can also arise when the underlying model is poor because: (a) the model family cannot represent the concept well, such as linear separators approximating circular regions, or (b) the model family is sufficiently expressive but the model itself is not a good fit. Agnostic Learning (Kearns et al., 1994) addresses (a) by defining the minimal error rate of any classifier in a given family $\mathcal{G}$ of classifiers $g : \mathcal{X} \to \{-, +\}$:

$$\text{opt}(\mathcal{G}) := \min_{g \in \mathcal{G}} \Pr_{x \sim D}[g(x) \neq f(x)] \in [0, 1].$$

If $\mathrm{opt}(\mathcal{G})$ is large, then any classifier in $\mathcal{G}$ will have high misclassification rate. In our case, given a language model $\hat{p}_\theta$ parameterized by $\theta \in \Theta$, consider the family of thresholded-language-model classifiers:

$$\mathcal{G} := \{g_{\theta,t} \mid \theta \in \Theta, t \in [0,1]\}, \text{ where } g_{\theta,t}(c,r) := \begin{cases} + & \text{if } \hat{p}_\theta(r \mid c) > t, \\ - & \text{if } \hat{p}_\theta(r \mid c) \leq t. \end{cases}$$

It follows immediately from Theorem 1 that

$$\mathrm{err} \geq 2 \cdot \mathrm{opt}(\mathcal{G}) - \frac{\max_c |\mathcal{V}_c|}{\min_c |\mathcal{E}_c|} - \delta.$$

When exactly one correct response exists per context (i.e., standard multiple choice, without IDK), the calibration term can be removed and bounds can be achieved even for $C = 2$ choices.

Theorem 3 (Pure multiple-choice)**.** *Suppose $|\mathcal{V}_c| = 1$ for all $c \in \mathcal{C}$ and let $C = \min_c |\mathcal{E}_c| + 1$ be the number of choices. Then,*

$$\mathrm{err} \geq 2\left(1 - \frac{1}{C}\right) \cdot \mathrm{opt}(\mathcal{G})$$

To illustrate, consider the classic trigram language model where each word was predicted based only on the prior two words, i.e., a context window of just two words. Trigram models were dominant in the 1980s and 1990s. Trigram models, however, regularly output ungrammatical sentences. Consider the following prompts and responses:

$c_1 =$ She lost it and was completely out of...	$c_2 =$ He lost it and was completely out of...
$r_1 =$ her mind.	$r_2 =$ his mind.

Here, $V_{c_1} := E_{c_2} := \{r_1\}$ and $V_{c_2} := E_{c_1} := \{r_2\}$.

Corollary 2. *Let μ be uniform over $\{c_1, c_2\}$. Then any trigram model must have a generation error rate of at least 1/2.*

This follows from Theorem 3 because $C = 2$ and $\mathrm{opt}(\mathcal{G}) = 1/2$ for trigram models. The proofs of Theorem 3 and Corollary 2 are in Section C. Although n-gram models can capture longer-range dependencies for larger n, data requirements scale exponentially in n.

We now revisit the letter-counting example from the introduction. To see that this is a poor model issue, note that the DeepSeek-R1 reasoning model reliably counts letters, e.g., producing a 377-chain-of-thought that includes:

> Let me spell it out: D-E-E-P-S-E-E-K.
> First letter: D — that's one D. Second letter: E — not D. Third letter: E — not D...
> So, the number of Ds is 1.

Assuming similar training data, this suggests that R1 is a better model for the task than the DeepSeek-V3 model. One representational challenge that reasoning overcomes is that modern language models represent prompts by *tokens*, e.g., D/EEP/SEE/K, rather than individual characters (DeepSeek-AI et al., 2025).

3.4 Additional factors

Errors may occur due to a combination of multiple factors, including the ones discussed above and several others. Here, we highlight a few.

- **Computational Hardness**. No algorithm run on a classical computer, even an AI with superhuman capabilities, can violate the laws of computational complexity theory. Indeed, AI systems have been found to err on computationally hard problems (Xu et al., 2024). Observation 2 of Section D illustrates how Theorem 1 applies to intractable queries of the form *"What is the decryption of c?"* and IDK is a valid answer.
- **Distribution shift**. A well-known challenge in binary classification is that training and test data distributions often diverge (Quiñonero-Candela et al., 2009; Moreno-Torres et al., 2012). Analogously, errors in language models often stem from out-of-distribution (OOD) prompts that differ substantially from the training distribution. A question such as, "What's heavier, a pound of feathers or a pound of lead?" may be unlikely in the training data and may induce erroneous answers in certain models. Similarly, distribution shift could be a factor in the letter-counting example above, though the fact that reasoning models correctly count letters suggests that the poor models may be a greater factor.
- **GIGO: Garbage in, Garbage out**. Large training corpora often contain numerous factual errors, which may be replicated by base models. The statistical similarity of GIGO for both classification and pretraining is self evident, and hence we do not provide a formal treatment. However, it is important to recognize GIGO among statistical factors, as language models have been shown to replicate errors from training data (Lin et al., 2022b; Levy et al., 2021; Alber et al., 2025).

GIGO also offers a natural segue to the topic of post-training, which decreases certain GIGO errors, such as common misconceptions and conspiracy theories (Ouyang et al., 2022; OpenAI, 2023a; Costello et al., 2024). The next section explains why some hallucinations persist—and may even be exacerbated—by current post-training pipelines.

4 Post-training and hallucination

Post-training should shift the model from one which is trained like an autocomplete model to one which does not output confident falsehoods (except when appropriate, e.g., when asked to produce fiction). However, we claim that further reduction of hallucinations is an uphill battle, since existing benchmarks and leaderboards reinforce certain types of hallucination. We therefore discuss how to stop this reinforcement. This is a socio-technical problem in the sense that, not only do the existing evaluations need to be modified, but these changes need to be adopted in the influential leaderboards.

4.1 How evaluations reinforce hallucination

Binary evaluations of language models impose a false right-wrong dichotomy, award no credit to answers that express uncertainty, omit dubious details, or request clarification. Such metrics, including accuracy and pass rate, remain the field's prevailing norm, as argued below. Under binary